Two markets, one pricing pipeline

IAA AI-Assisted Hackathon 2026: final write-up

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28 October 2026

	Week 1: developed market	Week 2: emerging market
Data	2021–24, 163,150 claims; 626,639 rows to price	2023–24, 9,258 claims; 209,063 rows to price
Submitted file	six-expert geometric mean, recalibrated per line	v13: ensemble ranking with segment levels read from the leaderboard
Public / private D^2	0.12851 / **0.13830**	**0.17172** / pending (v12, public 0.17095, also selected)

The score is the Tweedie deviance D^2 with power 1.5 over all rows of the three lines. My first Week 2 file, a plain LightGBM, scored 0.12519. Three ideas carried the work: get the target right before modelling; in a new market, start from what the old market knows and learn only the differences; and keep the level of a segment apart from the ranking inside it, because the 2025 book is too unlike the training years for a back-test to judge levels.

1 Data and target

Each week runs the same three scripts (read, clean, build features), each with an executed notebook as evidence. Every fix is logged with the rows it changed. No row is dropped and nothing is imputed. Among the issues found: customer ids that had lost their leading zeros (123,664 rows); exposure recorded in days or on a 360-day basis (13,880 + 595 rows); sums insured in units instead of thousands (3,527 + 349); a case reserve typed with an extra digit (11.9m for 1.19m); impossible years licensed, restored from the same policy's other years (3,717); and 604 claims stored twice under a second id, which inflated 2022 claim counts by 1.4%.

Target. The ultimate cost of the claims whose loss date falls in the row's period, net of excess and recoveries, repudiated claims excluded. Claims are valued at 31 December 2024, so 2024 is not fully reported (about half of its bodily-injury cost is in). Every model's offset is log(exposure × share of claims reported), from a reporting pattern per peril. Open claims are developed with the recovery ratio of closed ones, costs are trended to 2025 per peril, and costs above the 99.5% quantile are capped and returned as a load per line (1.02 to 1.06).

Feature selection. A column is kept only if it beats 20 random "probe" columns out of time, alone and inside a small LightGBM, with a pass mark that lets at most 1 in 20 noise columns through. Week 1 keeps 53 / 44 / 49 features (motor / buildings / contents), Week 2 only 28 / 14 / 9: two years of a small book cannot prove weaker effects.

2 Week 1: six experts and a geometric mean

For each line, six experts are each trained three times: on 2021–22 to price 2023, on 2021–23 to price 2024, and on every year to price 2025. All are LightGBM with an exact log offset, except the GLM.

Expert	What it models
`total`	Tweedie GBM on capped total cost, times the line's large-loss load
`glm`	Tweedie GLM on banded numeric factors and grouped categories
`peril`	one Tweedie GBM per peril, with its own reporting share and load, summed
`fs_gamma`, `fs_lognorm`	Poisson frequency × claim size (Gamma; or lognormal with a risk-specific variance)
`fs_peril`	frequency × claim size for each peril, summed

The predictions are combined row by row with an equal-weight geometric mean, then recalibrated per line with a two-parameter Tweedie GLM, $\log E = \log(\text{exposure} \cdot L) + b_0 + b_1 \log(\text{geo}\ /L)$, where L is the line's mean rate. It is fitted on the two back-test years (b_1 = 1.04 motor, 1.10 buildings, 1.09 contents).

Back-test D², all lines (calibration fitted on the other year)	2023	2024
one LightGBM on total cost	0.1364	0.1338
best single expert (`fs_peril`)	0.1379	0.1365
geometric mean of the six, recalibrated	**0.1386**	**0.1375**

Frequency × claim-size experts beat a single cost model in every line and year, and the six together beat each of them. A seventh expert (a claim-size mixture of experts) gave the same private score, so I kept the simpler file. The table comes from the rebuilt pipeline. The submitted file came from the 22 October code, which reported 0.1380 / 0.1367 and predates the duplicate-claim and years-licensed fixes. Its 2025 predictions have a log correlation of 0.996 to 0.998 with the rebuild.

3 Week 2: adapting to the emerging market

The new market broke most of what Week 1 could take for granted:

	Week 1	Week 2
History	4 years, 163k claims	2 years, 9k claims (about 1,000 per home line)
New business in the year priced	close to the training mix	77% of 2025, against 12–15% of training rows
Gender	not a rating factor	a permitted rating factor
Weather of the year	inside the training range	2025 far outside it (rainfall anomaly −128 mm against ±3 mm)
Buildings experience	stable	2023 fire-cost spike: 2024 came in at 0.59 of 2023's prediction

Transfer from the developed market. A LightGBM is fitted on every Week 1 row with only the factors that mean the same thing in both markets (30 motor, 24 buildings, 26 contents), always including renewal count, and its log rate goes into the offset of the Week 2 models. The local trees then learn only how this market differs, and need just 20 to 110 trees. Each Week 1 expert is rebuilt on a prior of its own component (frequency, claim size, each peril). A peril with fewer than 50 claims (300 for claim sizes) uses its prior alone, re-levelled. This-year weather is dropped.

Choices made by cross-validation (five folds of customers over 2023–24, since two years give only one forward test): one-split trees for motor and contents; a strong GLM ridge (a weak one overfits buildings below D² = 0); a member set per line instead of all six (motor `total` + `fs_peril` + `glm`, buildings `fs_lognorm` + `peril`, contents `total` + `glm`), because two years hold too few claims to learn claim sizes; and a recalibration slope fixed at 1, since a fitted slope did not carry from one year to the other. New-business and gender factors are measured on out-of-fold predictions and shrunk towards no effect.

Forward test: fit on 2023, score 2024, all lines	D²
LightGBM on Week 2 data only	0.1026
developed-market prior alone	0.1259
LightGBM on top of the prior	0.1277
ensemble on priors	**0.1300**

4 What the leaderboard taught me

The ensemble (v4) scored 0.16198 in public, below my earlier single transfer LightGBM (0.16529), although it ranked the 2024 risks better. The difference was in the levels. The 2024 back-test is 13% new business and 2025 is 77%, so the back-test can judge the ranking inside a segment but not the level of new business, buildings or gender.

So I split the problem. **Ranking.** v5 keeps the earlier file's total in each of ten segments (line × new business × gender) and replaces the ranking inside them with a 50/50 geometric blend with my ensemble. On 2024 rows weighted to the 2025 mix it ranks better (within-segment D^2 0.1067 against 0.1016). It scored exactly 0.16529: the public score barely reacts to ranking inside segments.

Levels. For Tweedie p = 1.5, multiplying a segment's predictions by c changes the deviance by exactly

$$\Delta D = 4\left[(c^{-1/2}-1)A + (c^{1/2}-1)B\right], \qquad A = \sum y\,\mu^{-1/2}, \quad B = \sum \mu^{1/2},$$

so the best level is $c^* = A/B$, and any scored file that differs from another only by segment factors measures it. One model, linear in the ten best levels, fits 13 public scores within 0.0001. Five of them, left out of the fit, were predicted up to one constant offset. It showed that motor needed a much wider gender split than any model of 2023–24 gave: new-business men about ×1.37 on v5, women about ×0.97 (Figure 2).

Luck. The public score covers only a sample of 2025 rows, so a segment's best level on it is off by $\mathrm{sd} = \sqrt{\phi/(\mathrm{share}\cdot\sum \mu^{1/2})}$: 8% for motor new-business men, 13% for buildings new business and 24% for buildings renewals at a 30% public share. The model fits that luck separately and gives two answers: the best levels for the public rows (v13) and the best levels for rows not yet scored (v12), which pulls the noisy segments back towards v5.

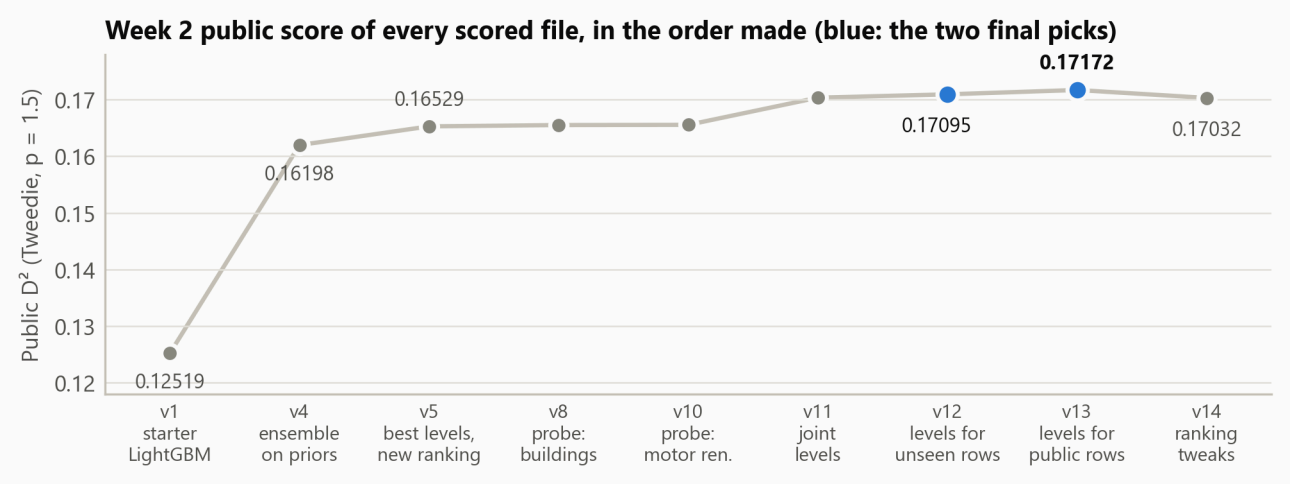


Figure 1: Every scored Week 2 file. Levels moved the public score; ranking changes (v5, v14) hardly did.

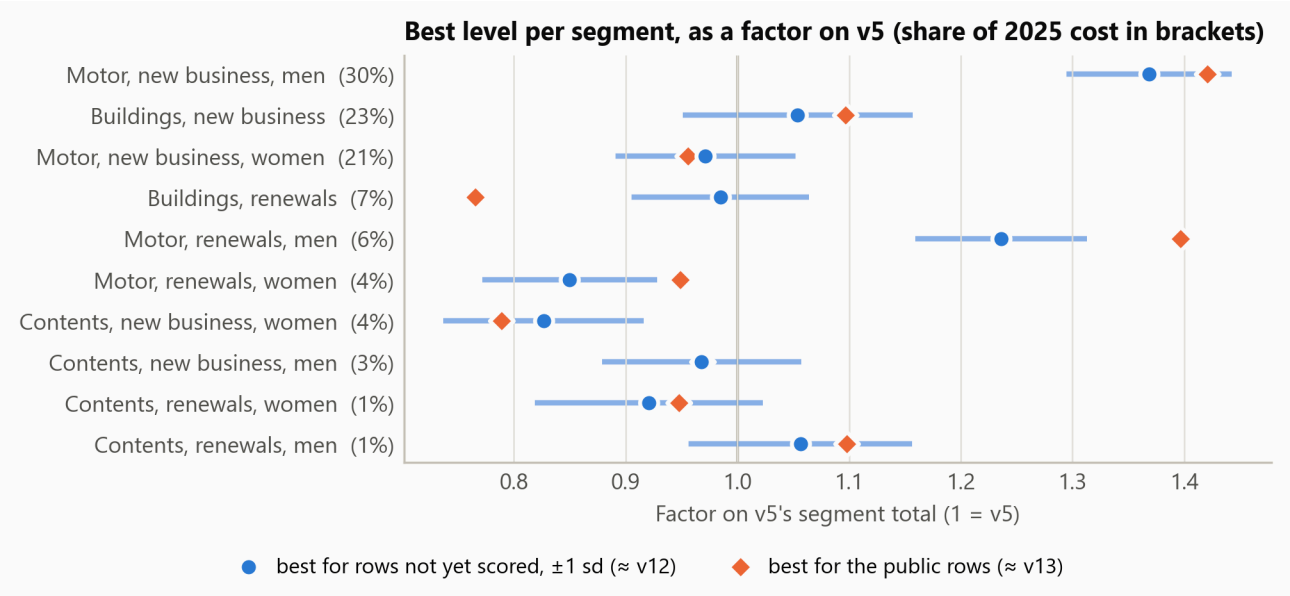


Figure 2: Best level per segment. v12 targets the blue points, v13 the orange ones. Where they differ most (buildings renewals, motor renewal men), the public rows are least informative.

v12 scored 0.17095 (expected 0.1708) and v13 0.17172 (expected 0.1714). Against v5, the model expects v12 to gain 0.0042 on unseen rows and v13 0.0037 (sd about 0.0015). On Kaggle I selected both: v12 is my best estimate for the final ranking, and v13 wins only if the true levels are as extreme as the public rows suggest. The Week 2 file submitted here is v13, my best score. Two

last ranking changes (no this-year weather in the buildings and contents model, motor spread ×1.05) scored 0.17032 (v14): 0.0006 below v12, with a public-luck sd of 0.0008. The leaderboard cannot read that either way, so v14 was not selected.

In short, what the new market changed. (1) I stopped learning everything locally: the developed market supplies the relativities and the emerging data only corrects them. (2) Simpler local models won: one-split trees, a strong ridge, fewer experts, no fitted slope. (3) The back-test was no longer enough, because 2025 is mostly new business, so ranking and level were judged on different evidence. (4) The public leaderboard became a noisy measurement to model, not a score to maximise.

5 Major decisions

Decision	Rejected	Evidence
Fix data errors with evidence; drop and impute nothing	dropping or imputing rows	cleaning log per round
Geometric mean of experts, recalibrated per line (Week 1)	one LightGBM	+0.0022 (2023) and +0.0037 (2024) back-test D^2
Developed-market prior in the offset (Week 2)	Week 2-only models	+0.025 forward D^2
Members per line by cross-validation; slope fixed at 1	all six experts; fitted slope	contents: all six 0.1071 against `total` alone 0.1087
Segment levels kept apart from ranking	trusting the back-test's levels	v4 0.16198 against v5 0.16529
Aim at unseen rows, not the public sample	chasing the public score	public luck sd 8–24% per segment

6 Use of AI

I worked alone, with an AI coding agent (Claude Code in VS Code) as the partner at every step. The agent wrote and ran most of the code and drafted this write-up. The questions, the direction and every decision to implement stayed with me.

- **Prompting to explore ideas.** Most of the work was a long series of prompts that proposed and tested ideas: the developed-market prior, the expert ensembles, segment-level probes, the public-luck model. Each idea became a script and was measured out of time before it counted.
- **Context to point the search.** Each prompt came with context: the case study, what had already been tried and why it failed (kept in `KNOWLEDGE.md`), and the question to answer. That kept the agent exploring in the direction I chose instead of starting over or repeating dead ends.
- **Skills for coding and best practice.** Two project skills in the repository (`.claude/skills/data-scientist`, `ml-engineer`) set how the agent works: frame the problem first, test out of time, distrust its own gains, keep code reproducible and versioned.
- **Every idea questioned independently.** A result was not accepted from the run that produced it. Each idea was questioned separately, so the agent had to check its own work. Factor searches looked strong when the same two years chose and scored them; a permutation test then showed real factors pass no more often than shuffled ones, so only gender was kept. A review after the Week 1 deadline found a sign error in the contents frequency trend; it is documented, kept in the code that reproduces the submitted file, and fixed in the rebuild.
- **Human sign-off before implementation.** Nothing went into the pipeline or a submission until I had reviewed the plan and its evidence. I chose the target definition, what evidence counts, which files to submit and select, and when to stop probing the leaderboard.

7 Assumptions

- 2025 claims follow 2023–24 after trend (motor severity 0–7.9% a year by peril, buildings severity 6%, contents frequency 1.4% and severity 3.4%). Reporting patterns are stable, and open claims develop like closed ones.
- For factors defined the same way in both markets, the developed market's relativities are a sound starting point. The emerging market differs in level and in a few effects that local models can learn.
- Public-leaderboard rows are a random sample of 2025 (about 30%). The order of v12 and v13 holds for any share from 20% to 50%.
- Gender is a permitted rating factor in the emerging market, as the case study states.

8 Limitations

- Two years of emerging-market data, and buildings is lumpy: a few large claims decide its public score, and its levels carry 8–10% uncertainty.
- Week 2 levels rest partly on the public leaderboard. That is allowed in the competition, but in practice it amounts to calibrating on a sample of the book's own outcomes, and the bands in Figure 2 are wide.

- Back-tests cannot judge new-business levels, and ranking changes such as v14 are smaller than the public score's noise.
- The submitted Week 1 file keeps the contents trend sign error. The per-line recalibration absorbs most of it (intercept about +10%), but contents relativities may be slightly off.

9 Guardrails

- **Leakage.** Outcome-driven columns are excluded: `renewal_decision`, and `customer_service_contacts` (zero on 0.2% of rows with a claim, against 67% without). Claims history is built as known on 1 January of each year.
- **Governance.** The written premium is excluded (a model using it would copy the old tariff), as are marketing and service fields and personal data such as surname. Gender is used only where permitted, and its factor moves price between men and women without changing the line's total.
- **Evidence before every submission.** Each file is tested out of time first, then registered as a new version with its evidence, description and checksum. Versions are never overwritten.
- **File checks.** A file is saved only with one row per sample row id, in order, every prediction finite and positive. The Week 1 code adds block and escalate rules: back-test $D^2 \le 0$ blocks; A/E outside $1 \pm 5\%$, feature drift (PSI ≥ 0.25) or a pure-premium shift above 15% against history goes to actuarial sign-off.
- **Noise control.** Probe columns in feature selection, permutation tests on factor searches, and the public-luck model all guard against fitting noise, including noise an AI agent would report as a gain.
- **Reproducibility.** Every step is a script under git. The data steps give the same bytes on every run, and the Week 1 file rebuilds from cached expert predictions in a minute (largest relative difference 2e-15).

Appendix (outside the five graded pages)

A. Week 2 submission register (from `2nd round/submissions/versions.csv`).

Version	Public D^2	What it is
v1	0.12519	starter: one LightGBM per line, no prior, no trend or large-loss treatment
v2	not scored	LightGBM on the developed prior, with new-business and gender factors
v3	not scored	ensemble on developed priors, recalibration slope fitted (kept for comparison)
v4	0.16198	ensemble on developed priors, level only (slope 1)
v5	0.16529	earlier best file's segment totals; ranking 50% that file, 50% ensemble
v6	not scored	as v5, ranking 25% / 75%
v7	not scored	probe: v5 with motor new business x1.15
v8	0.16552	probe: v5 with buildings new business x1.08
v9	not scored	probe: v5 with contents new business x1.10
v10	0.16556	probe: v5 with motor renewals x1.07
v11	0.17036	v5 with every segment at its best level from all scores
v12	0.17095	v5 with every segment at its best level for rows not yet scored
v13	0.17172	v5 with every segment at its best level for the public rows (Week 2 file)
v14	0.17032	v12's levels; buildings and contents ranked without this-year weather; motor spread x1.05
v15	not scored	v12 with the motor spread x1.05 only

B. Segment levels (from `2nd round/reports/11_joint/levels.csv`; factors on v5).

Segment	Share of 2025 cost	Public-row luck sd	Best for public rows	Best for unseen rows	sd
Motor, new business, men	30.2%	8.3%	1.421	1.369	0.074
Buildings, new business	23.3%	12.8%	1.097	1.054	0.103
Motor, new business, women	21.5%	9.0%	0.956	0.971	0.081
Buildings, renewals	6.7%	23.5%	0.765	0.985	0.079
Motor, renewals, men	5.8%	16.8%	1.397	1.236	0.077
Motor, renewals, women	4.1%	18.2%	0.949	0.850	0.078
Contents, new business, women	3.8%	9.8%	0.789	0.826	0.090
Contents, new business, men	3.1%	10.2%	0.969	0.968	0.089
Contents, renewals, women	0.9%	18.6%	0.948	0.921	0.102
Contents, renewals, men	0.8%	19.5%	1.098	1.057	0.100

C. Files and how to reproduce. The two prediction files, their checksums and sources are in `final/1_predictions/manifest.csv`. The code is at https://github.com/IKSHRESTHA/IAA-Hackathon (map in `final/4_code/README.md`). `python final/build_final.py` copies and checks the prediction files, writes a code snapshot from git, and re-renders this document.